Sahaj Bhadja

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Summary

Software Engineer with 2+ years in **systems-level programming** and **robotics-adjacent workflows**, building reliable **C/C++ and Python** components, validating runtime behavior, and debugging cross-boundary issues across software, data, and hardware-like interfaces.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **Python-based simulation pipelines** for autonomous systems, processing **100K+ trajectory logs** to evaluate control behavior and improve long-horizon planning performance by **18%**.
- Debugged low-level runtime anomalies across model/system boundaries and produced repeatable validation traces for cross-functional hardware/software investigations.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered cross-platform service workflows with **OOP**, improving response time by **30%**; maintained robust CI validation across 3 OS environments for deterministic behavior.
- Owned **20+ unit/integration test suites** (Jenkins, PyTest, Mockito), strengthening defect isolation and reliability under automated release conditions.

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built reliability-focused backend components for high-throughput systems, reducing latency from **1.2s to 350ms** and improving deterministic processing behavior through query/runtime tuning.
- Implemented **validation and release checks** for 5K+ monthly jobs, improving system robustness, fault detection, and operational maintainability.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Implemented modular service logic and request processing controls for **10K+ daily requests**, improving stability of production workflows under variable load.
- Strengthened release confidence through CI/CD automation and targeted debugging workflows, reducing turnaround time by **60%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Developed API/WebSocket modules supporting 8+ workflows and 2M+ synchronized records with structured logging for runtime diagnostics and integration verification.
- Automated deployment/testing procedures, reducing deployment time by **50%** while increasing consistency of system bring-up and validation checks.

Skills

Languages: **C++**, **Python**, Java, SQL, JavaScript/TypeScript
Embedded/Systems: Embedded Linux familiarity, systems architecture, runtime debugging, low-level issue triage
Robotics & Controls: simulation-based validation, autonomous behavior testing, sensor-data workflow familiarity, control tuning exposure
Interfaces: UART/SPI/I2C familiarity, communication pipeline reasoning, API integration
Tooling: Jenkins, PyTest, Mockito, Docker, GitHub Actions, Git, CI/CD, release diagnostics
Engineering: cross-functional collaboration with mechanical/electrical peers, documentation, verification-first development

Projects

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented **6 scheduling algorithms** and runtime visualizations across **1K+ process combinations**, improving systems-level debugging and deterministic execution reasoning.

CaptionCraft – Image Caption Generator Live | *Transformers, PyTorch, Flask*
- Built and validated inference workflows over a **10K image / 100K caption** dataset, improving accuracy by **15%** through repeatable model evaluation pipelines.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Implemented API-driven operational workflows with monitoring-backed controls, supporting **100+ daily appointments** and improving record retrieval by **2x**.